

Agent-Assisted Data Construction for Industrial Visual Inspection: Toward Automated Anomaly Detection

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Abstract

Industrial anomaly detection (IAD) is moving beyond distinguishing normal from defective samples toward identifying the defect type, describing it, and localizing the defect region, leveraging the visual and semantic knowledge that large vision-language models (LVLM) learn from large-scale image-text pretraining. However, the LVLM lacks the inspection-specific knowledge needed to identify subtle industrial defects. Prior work addresses this through prompt-based zero-shot methods, which cannot fundamentally improve domain performance, and supervised fine-tuning on curated QA datasets, which imposes substantial human cost for sample-level defect annotation, QA construction, hard-negative design, rationale writing, and quality verification. To overcome this bottleneck, we propose the *Agent-Assisted Data Construction Framework*, which redefines expert input from repeated sample-level annotation to providing seed domain knowledge and reviewing agent outputs. A VLM-based Visual Evidence Inspector extracts a VEI evidence card — a structured summary of the defect region, visible cues, and related seed domain knowledge. An LLM-based Challenger converts the card into MMAD-style QA candidates, and a VLM-based Critic checks whether each candidate is supported by visual evidence in the input image and consistent with the seed domain knowledge. Accepted candidates are exported as SFT/DPO training rows that experts can inspect, trace, and correct, while low-quality candidates are filtered or routed for repair. By automating the construction of training data grounded in visible evidence, the framework provides a scalable path toward automated anomaly detection.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing; Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

vision-language models, industrial anomaly detection, multi-agent framework, data construction, visual question answering

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1 Introduction

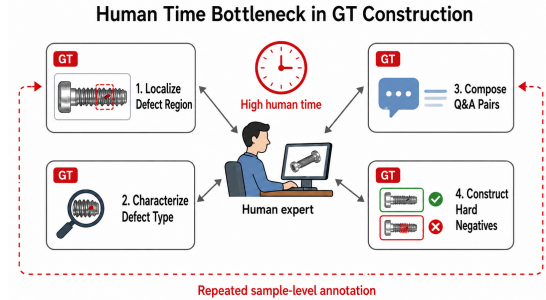


Figure 1: Expert annotation becomes a bottleneck because experts repeatedly localize defect regions, identify defect types, write QA pairs, and design hard negatives for each sample. Our framework reduces this repeated sample-level burden by assigning these tasks to specialized agents.

Industrial anomaly detection (IAD) has progressed beyond distinguishing normal from defective samples toward defect-centered reasoning that involves identifying the defect type, localizing the defect region, and describing supporting visual evidence. Large vision-language models (LVLM) bring visual and semantic knowledge learned from large-scale image-text pretraining that aligns naturally with these tasks, enabling models to answer questions about defect category, location, abnormal cues, and object-level implications. Benchmarks such as MMAD [1] formulate this LVLM-IAD setting through multiple subtasks, including anomaly discrimination, defect classification, defect localization, defect description, defect analysis, object classification, and object analysis.

However, the LVLM lacks the inspection-specific knowledge needed to identify subtle industrial defects. Recent LVLM-IAD methods such as AnomalyCoT [2], M3-AD [3], JUDO [4], and Echo [6]

Table 1: Repeated expert work across LVLM-IAD datasets and training-based methods. O = expert work used directly, partially, or indirectly, or applied through existing annotation or external expert models within the framework. X = expert work not performed by the framework, irrelevant to the framework’s purpose, or, in Ours, automated by a specialized agent. In our framework, X-marked rows are handled by VEI (defect-region evidence and sample-level annotation support), Challenger (QA format and hard negatives), VEI+Challenger (reasoning), and Critic (data quality review); O-marked rows remain as expert roles.

Expert work	MMAD [1]	AnomalyCoT [2]	M3-AD [3]	JUDO [4]	AgentIAD [5]	Ours (this work)
Specify defect region	O	O	O	O	O	X
Provide sample-level defect annotation	O	O	O	O	O	X
Design QA / output format and answer	O	O	O	O	O	X
Design distractors / hard negatives	O	O	X	O	X	X
Provide reasoning / CoT / rationale	O	O	O	O	O	X
Review generated data quality	O	O	O	X	X	X
Define reward / verifier / policy rules	O	O	O	O	O	O
Provide seed domain knowledge	O	O	O	O	O	O

extend industrial inspection toward visual question answering, defect description, and multimodal reasoning, while training-free approaches such as WinCLIP [7] and LogSAD [8] provide vision-language alignment without model updates. Training-free and multi-expert methods [5, 9] do not fundamentally improve domain-specific IAD performance, leaving supervised fine-tuning as a practical direction for teaching the LVLM industrial inspection knowledge.

Yet existing supervised pipelines impose substantial human cost: AnomalyCoT alone reports 230 expert-hours from 10 annotators for defect-region annotation, defect-type labeling, QA construction, hard-negative design, rationale writing, and quality verification, and this sample-level burden does not scale to new objects, defect types, or inspection environments.

To overcome this bottleneck, we propose an annotation-efficient multi-agent framework for LVLM-IAD data construction. Instead of asking experts to provide detailed defect annotations for every sample, our framework shifts expert input toward basic normal/anomaly distinction and object-specific seed domain knowledge. A VLM-based Visual Evidence Inspector (VEI) extracts a VEI evidence card — a structured summary of the defect region, visible cues, and related seed domain knowledge. An LLM-based Challenger converts the card into MMAD-style QA candidates, and a VLM-based Critic [10] verifies them through a Critic verification step that checks whether each candidate is supported by visual evidence in the input image and consistent with the seed domain knowledge. Accepted candidates are exported as SFT/DPO training rows that experts can inspect, trace, and correct, while low-quality candidates are filtered or routed for repair. By automating the construction of training data grounded in visible evidence, the framework provides training data that can be expanded to new objects and defect types with less repeated expert annotation, bringing the field closer to automated anomaly detection.

The main contributions of this work are:

- We propose an agent-assisted multi-agent framework for LVLM-IAD training data construction that replaces repeated per-sample expert annotation with three specialized roles — Visual Evidence Extraction (VEI), QA Candidate Generation (Challenger), and Quality Verification (Critic).
- We design a seed-domain-knowledge-driven SFT/DPO data construction mechanism in which VEI, Challenger, and Critic

use the same seed knowledge differently according to their roles, enabling the generation of SFT/DPO-oriented QA candidates and hard negatives.

- We verify our framework on the MMAD benchmark and observe stable improvements with the largest gain in defect localization, indicating that VEI evidence and Challenger-generated QA candidates help the model learn defect-localization cues.

2 Related Work

Industrial anomaly detection (IAD) has progressed from binary detection toward defect-centered reasoning. Classical methods such as PatchCore [11] model the feature distribution of normal samples, and recent zero-shot approaches including WinCLIP [7] and LogSAD [8] leverage vision-language foundation models for training-free detection. However, these methods do not fully address higher-level inspection questions such as defect type, cause, or supporting visual evidence. To bridge this gap, LVLM-based IAD studies including AnomalyCoT [2], M3-AD [3], JUDO [4], and Echo [6] extend industrial inspection toward visual question answering, defect description, and multimodal reasoning. Yet they impose substantial expert annotation cost: for instance, AnomalyCoT alone reports 230 expert-hours from 10 annotators plus additional verification, and these pipelines often depend on sample-level defect-region annotation, defect-type labeling, QA design, and quality checking, without explicitly separating vision-grounded verification as a dedicated data-quality gate.

Unlike prior agent-based methods such as AgentIAD [5] and EAGLE [9], which use agents to help the model reason step by step or detect anomalies without any fine-tuning when the model makes predictions, our framework uses agents to construct SFT/DPO training rows that experts can inspect, trace, and correct, through specialized evidence extraction, QA generation, and visual-evidence verification. We support rather than remove expert involvement by assigning the sample-level burden to specialized agents: VEI for defect-region evidence extraction, Challenger for QA and hard-negative generation, and Critic for quality verification, while experts retain seed domain knowledge provision and set criteria for reviewing agent outputs. Table 1 compares eight expert-intervention areas across prior frameworks and ours, guiding our agent design.

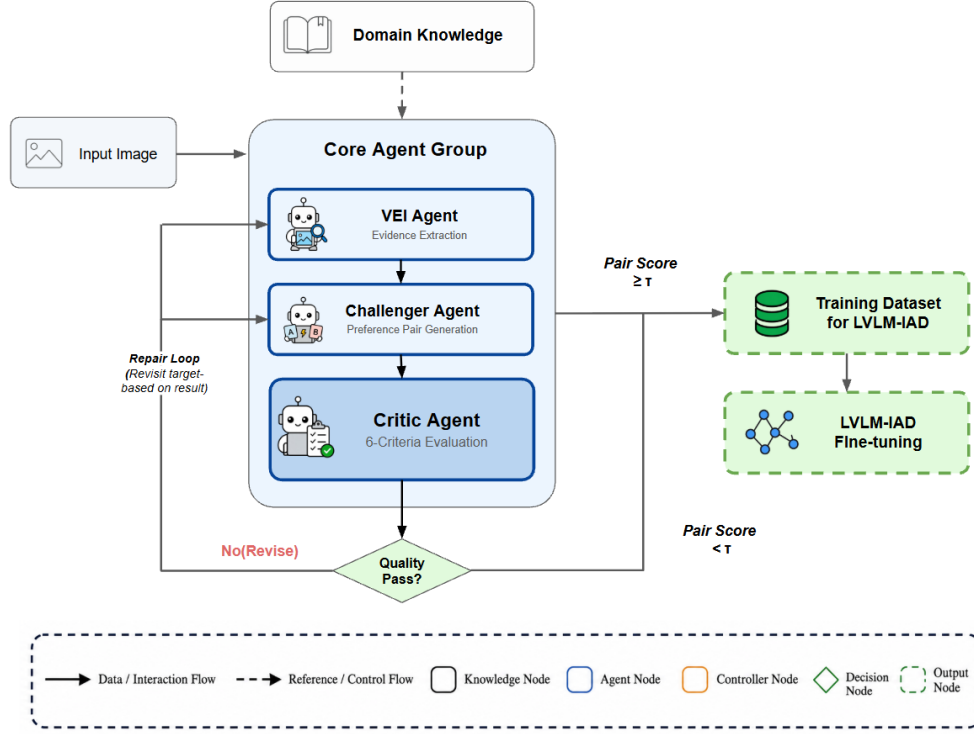


Figure 2: Agent-Assisted Data Construction Framework. Agents turn visual evidence into QA candidates, Critic Agent checks and routes repairs, and Compiler exports accepted SFT/DPO rows.

3 Framework

Figure 2 illustrates the Agent-Assisted Data Construction Framework for LVLM-IAD data construction. The framework is designed to convert image-level visual evidence into SFT/DPO training rows that experts can inspect, trace, and correct, through specialized agents. Table 2 summarizes the input and output of each agent.

3.1 Overall Framework

The framework converts an LVLM-IAD image into reviewable training data through four stages: visual evidence extraction, QA and hard-negative generation, quality verification, and final SFT/DPO row construction. Rather than generating training data in a single end-to-end step, the framework separates the process into specialized agent modules so that each intermediate artifact can be inspected and repaired.

At the framework level, the input consists of an LVLM-IAD image and object-specific seed domain knowledge. VEI first extracts visual evidence and summarizes it as an evidence card. Challenger then uses this evidence card to generate MMAD-style QA candidates, chosen answers, and hard negatives. Critic evaluates whether the generated outputs are supported by visual evidence and suitable for SFT/DPO training. Finally, Compiler — a deterministic, non-LVLM module — formats only Critic-approved candidates into SFT/DPO rows while preserving the linkage from each exported row back to its source image, evidence card, and Critic scores. Candidates marked for repair are routed back to VEI or Challenger rather than included in the export. The main design choice is transparency.

Table 2: Agent-level input and output.

Agent	Input	Output
VEI	Image views + seed domain knowledge	Evidence card
Challenger	Evidence card + seed domain knowledge	QA candidates + hard negatives
Critic	Image + evidence card + QA candidates	Scores + accept/reject + repair target
Compiler	Accepted candidates	SFT/DPO rows

Instead of hiding the full process inside one end-to-end model, the framework exposes the VEI evidence card, QA choices, chosen answers and hard negatives, critic scores, and repair target. Experts can then review, correct, and analyze failures rather than authoring every sample from scratch.

3.2 Agent Roles

Visual Evidence Inspector (VEI) Agent. VEI is the image-facing agent. It supports the expert task of deciding which defect region to inspect and which visual cues are present. It receives three views: the original image, a bbox-marked image, and an evidence crop. A TransFusion [12] anomaly map and Otsu thresholding are used to make these views before VEI runs. VEI then records structured observations about location, shape, color, texture, and structure. VEI does not make the final defect annotation, guess causes, or claim functional impact.

VEI gives Challenger a *VEI Evidence Card*. The card records the region, visible observations, seed domain knowledge match, forbidden claims, and fields needed by Challenger. Raw paths, raw defect keys, and crop metadata are stored in side logs. Failure signals include weak crop evidence, unclear object-part match, missing seed

domain knowledge match, or claims that need another VEI check. **Challenger Agent.** Challenger is the text-only agent. It turns the VEI evidence card into QA candidates. It generates questions, correct answers, hard negatives, and concise rationales. Challenger uses the VEI evidence card and seed domain knowledge. It does not reread the image or add new visual facts. The correct answer is an evidence-supported pseudo-label, not a final expert annotation.

From one VEI evidence card, Challenger generates seven MMAD-style questions: anomaly discrimination, defect classification, defect localization, defect description, defect analysis, object classification, and object analysis. Anomaly discrimination has two options. The other subtasks have four options. Challenger also generates an SFT target answer and a chosen answer for multimodal preference optimization [13, 14] from the locked correct option, visual evidence, and normality contrast [15]. It then selects a hard negative that looks reasonable for the same object but conflicts with the visual evidence [16]. Challenger logs why the chosen answer is supported and why the hard negative is inconsistent. Failure signals include invalid questions, unsupported answer claims, weak chosen/hard-negative gaps, trivial negatives, and hard negatives that do not conflict with the evidence.

Critic Agent. Critic Agent evaluates Challenger-generated chosen/rejected pairs for SFT/DPO training. The rejected answer is not a random wrong answer, but a hard negative that is domain-plausible while being inconsistent with the visual evidence. Critic therefore checks whether the chosen answer is vision-grounded and whether the rejected answer forms a meaningful fine-grained contrast. Each candidate is scored on six criteria: visual evidence discovery, evidence-based annotation inference, QA validity, hard-negative quality, rationale grounding, and domain-rule consistency. Each criterion uses a 0/0.5/1 anchor scale, and low-quality candidates are filtered or routed for repair.

4 Evaluation

We evaluate the proposed framework on MMAD, a benchmark for LVLM-based industrial anomaly detection [1]. MMAD includes seven subtasks: anomaly discrimination, defect classification, defect localization, defect description, defect analysis, object classification, and object analysis. During data construction, VEI, Challenger, and Critic run on Qwen-series 32B LVLM/LLM models to generate evidence cards, QA candidates, and verification decisions. The Critic-accepted SFT/DPO rows are then used to fine-tune Qwen2-VL-7B [17], which is evaluated on the seven MMAD subtasks.

Table 3: Per-subtask MMAD accuracy (%) of Qwen2-VL-7B (vanilla) and the proposed method (single representative seed). Δ is the change over vanilla.

MMAD Task	Qwen2-VL-7B	Proposed	Δ
1. Anomaly Discrimination	71.87	71.86	-0.01
2. Defect Classification	55.76	56.08	+0.32
3. Defect Localization	58.78	60.19	+1.41
4. Defect Description	64.96	64.98	+0.02
5. Defect Analysis	79.34	79.21	-0.13
6. Object Classification	92.51	92.50	-0.01
7. Object Analysis	84.62	84.62	0.00

As shown in Table 3, the proposed method improves the MMAD Average from 72.55 to 72.78, with the largest gain in defect localization (+1.41). Defect classification also improves by +0.32, while the remaining subtasks show only minor changes. This result is consistent with the framework design: VEI extracts defect-region evidence, and Challenger converts this evidence into QA and preference candidates.

Across five random seeds, the proposed method achieves an MMAD Average of 72.80 ± 0.03 , indicating that the improvement is stable rather than incidental. These results suggest that agent-generated visual evidence can provide useful supervision for LVLM-IAD training, especially for localization-related reasoning.

The current evaluation compares against the vanilla Qwen2-VL-7B baseline because our contribution focuses on data construction rather than model architecture or training strategy. Direct comparison with other trained LVLM-IAD methods under matched data budgets is left to future work. Future work will explore iterative agent-policy refinement, using Critic feedback to improve evidence extraction, QA generation, and verification quality across data construction rounds.

5 Conclusion

We presented the Agent-Assisted Data Construction Framework, which reduces repeated expert annotation in LVLM-IAD data construction by shifting expert input from sample-level annotation to providing object-specific seed domain knowledge and setting criteria for reviewing agent outputs. Our evaluation on MMAD shows a small but stable improvement with the largest gain in defect localization, suggesting that agent-generated visual evidence provides useful supervision for LVLM-IAD training.

The main design contribution is the structured division of data-construction tasks across specialized agents: VEI extracts defect-region evidence, Challenger generates QA candidates and hard negatives, Critic verifies chosen/rejected pairs, and Compiler exports only accepted rows. This design replaces fully manual sample-level annotation with a process in which agents generate candidate training data and experts review or correct the outputs, while preserving traceability and visual grounding.

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